BUSN5003

Assignment #1

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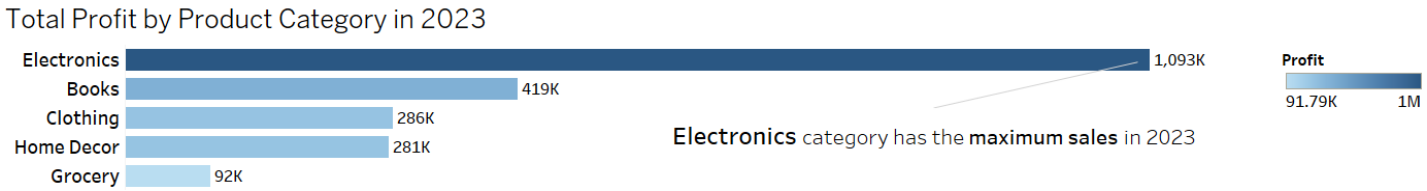
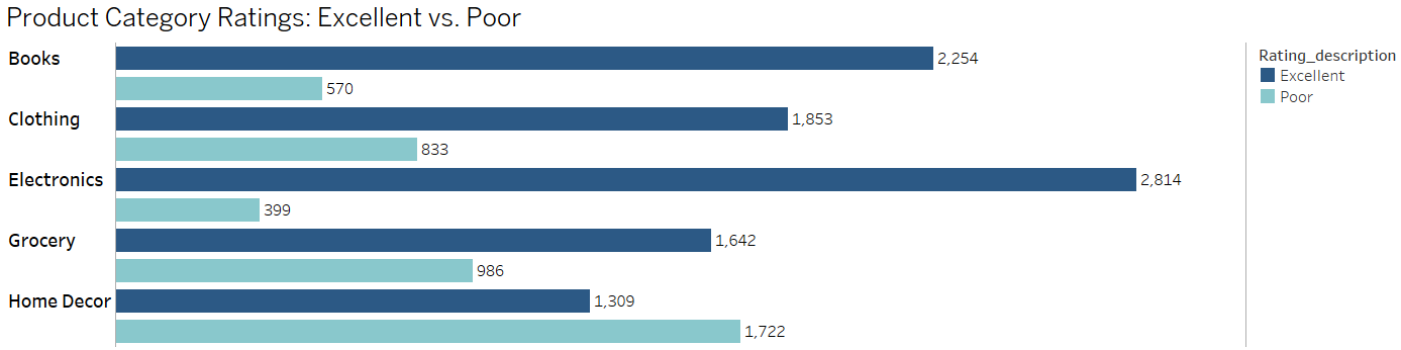
Insights

1. The relationship between the total profits and product ratings across the various categories.
2. Seasonal pattern in the profitability of various shipping methods.
3. The differences in shopping preferences across age groups.

The relationship between the total profits and product ratings across the various categories.

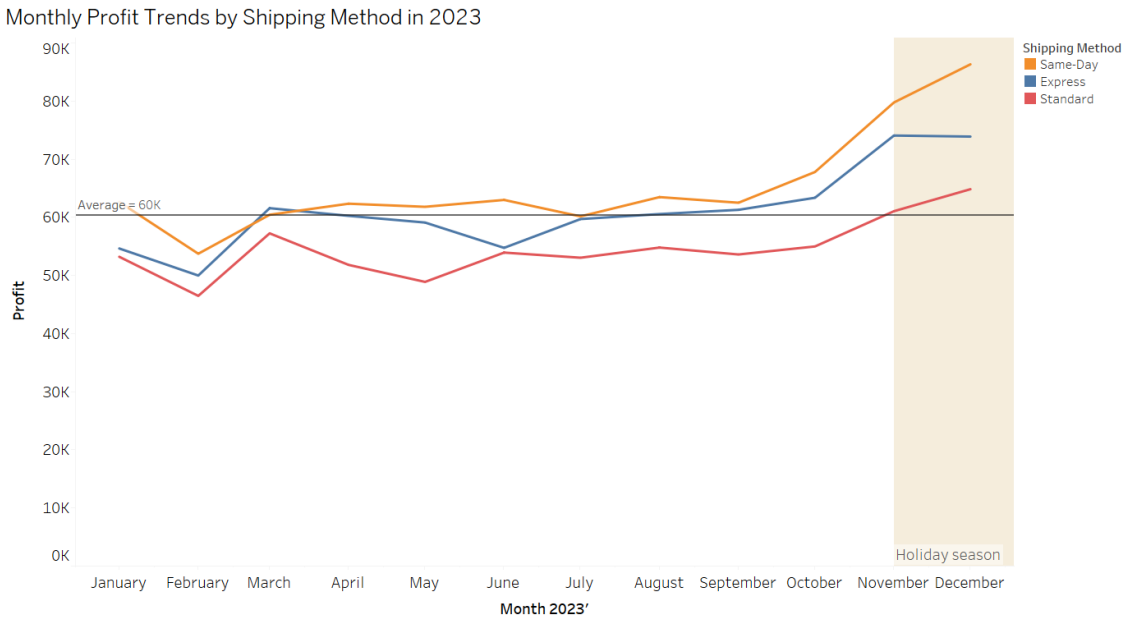




- **Electronics** stands out for generating the **highest profit** and **excellent ratings** which shows a **strong relationship** between profitability and customer satisfaction. This should be maintained by **quality assurance**, **customer support**, and more.
- **Home décor is relatively profitable** but it shows a **large** number of **poor ratings**, which needed to address the potential issues in product quality or understand the customer expectations.
- **Grocery** products generate **lower profits** but maintain a **balanced rating distribution**.

This insight reveals that **targeted strategies** are needed to enhance the profits and customer satisfaction within each product category.

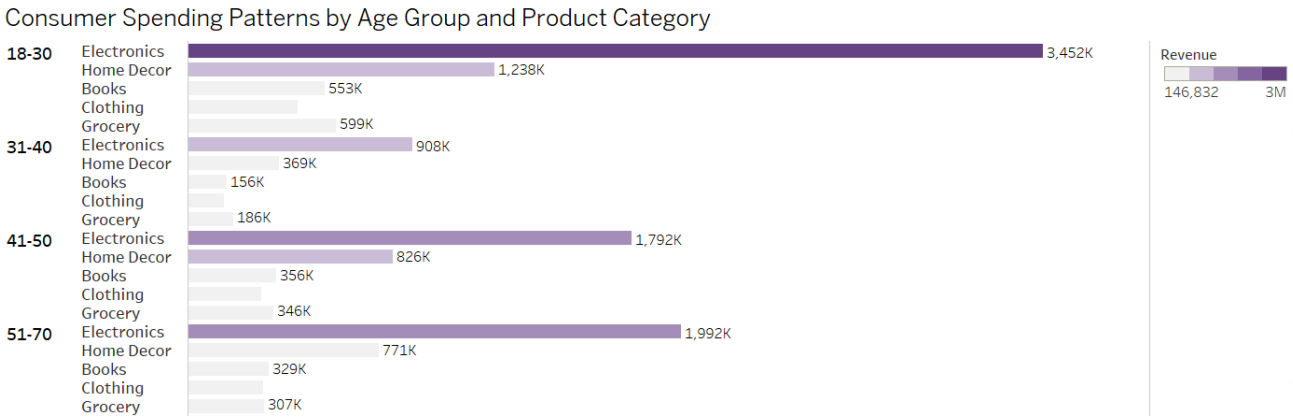
Seasonal pattern in the profitability of various shipping methods.



- From the chart **Same-day** and **express** shipping methods are frequently opted for and generate **higher profits**, especially during the **holiday seasons** in November and December.
- This shows that during **holiday shopping**, customers are willing to pay a **premium** for **'faster delivery'** which can enhance profits.
- On the other hand, **standard** shipping consistently yields **relatively lower profits** and crosses the average line at the end of the year. Therefore demand for standard shipping can be boosted by providing **discounts** and **offers** when opting for it.

These insights show the importance of **shipping strategies** with **seasonal consumer behavior** to maximise profits.

The differences in shopping preferences across age groups



- **Electronics** consistently **dominate** in all age groups. This shows that it is essential for marketers to maintain a **broad and inclusive approach** in related campaigns.
- Beyond the spending patterns in electronics, **younger consumers** (18 – 30) show significant spending on **groceries** (0.60 M) and **books** (0.55 M) which indicates an opportunity for the brands and retailers to **focus** on the younger consumers through **targeted campaigns**.
- **Groceries** has the most **consistent spending** across all groups, making it a stable marketing category.

This insight reveals the need for brands to adopt **age-specific marketing strategies** by focusing on younger groups through **social media campaigns**, **email newsletters**, and **value-driven offers** for older groups.

Visualisation/Decluttering Choice

Insight 2: Seasonal pattern in the profitability of various shipping methods.

Choice of visualisation: The line chart has been chosen to describe the trend of shipping method preferences. It was more suitable for displaying continuous time series and ideal to show changes over time.

For the line chart, the month of order date is used in the column, and the sum of profit in rows to present the monthly trend in profit. To show the trends in shipping methods, dragged the shipping methods to colour to differentiate the methods for easier comparison.

Decluttering Techniques:

- Added an average line to provide a measure for helping the viewers to quickly compare the performance with the overall average.
- Dragged a reference line and added a custom parameter to show November as the start of the holiday and filled the area above the line with a colour to highlight the period.
- To make the chart more readable adjusted the Y-axis interval to display values in thousands, edited the X- axis label, and gave a descriptive title to the chart to understand the meaning of the chart.
- Finally removed the gridlines on the rows for better clarity on shipping methods.

Data processing

Data preprocessing involved several key steps to prepare the data for analysis.

- Initially, duplicate records were checked but none were found.
- Declared data types appropriately:
  1. Age, units, and ratings are converted into numerical type with decimal precision reduced to 0
  2. Price, revenue, and profit were converted into numerical type with decimal precision reduced to 3 to minimize the clutter.
  3. Order item ID, order ID, and customer ID were converted to general type.
- Renamed the columns in Tableau for better clarity, for example: Product category → category.
- Null values were managed by filtering them out in Tableau, which retained most of the records.
- Created two calculated fields in Tableau for insight:
  1. Categorized ages into segments: 18 – 30, 31-40, 41-50, 51-70
  2. Converted numerical ratings to descriptive categories: excellent, very good, good, fair, and poor to enhance clarity.